



SafeRoute AI

Bengaluru NightRide - Predictive Road Safety Platform

PROJECT DOCUMENTATION

A simple, self-contained guide to what this project does, what problem it solves, how to set it up and run it, and the main tools used - written for non-technical readers.

IBM Bob National Hackathon 2026

Team 042 - Team Null Pointers

Problem Statement PS-3: RoadSafe India (Transport)

Version 1.0 - Documentation for project setup and execution



RED	Most dangerous - needs urgent fixing
ORANGE	Risky - high priority
YELLOW	Average - monitor and schedule work
GREEN	Comparatively safe



Screen Part	What It Shows and How to Use It
Colour-coded map (centre)	A satellite map of Bengaluru. Every road piece is drawn as a coloured line: red = most dangerous, orange = risky, yellow = average, green = comparatively safe. Use the + / - buttons to zoom, and the layers button to switch between map, satellite and traffic views.
Road list (left panel)	A searchable list of all 428 road pieces with their safety scores. Use the search box and the risk-level filter to narrow it down, or click a corridor name to see only that route.
Details panel (right)	Click any road piece on the map or in the list. The panel opens its Safety Report Card: the score, why it is risky (with percentage reasons), which people are most in danger, and the police-station accident history for that area.
What-If calculator (Simulate button)	Choose repairs such as new LED street lighting, speed cameras, pedestrian crossings or junction redesign. The system instantly recalculates the road's score and shows the expected improvement. You can save the scenario with a name for later review.
Top search bar (Ctrl + K)	Quick search across all road pieces and police stations using the keyboard.
Status badges (top right)	Show whether the background service is connected (Live) or stopped (Offline). If it says Offline, the maps and scores cannot load - see the troubleshooting note below.
Mobile screens	On a phone, the layout switches to tabbed views (Map / Roads / Details) so everything stays usable on small screens.

Step 1 - Install two free programs (only needed once)



Step 2 - Start the engine (background service)

Open the Command Prompt, go to the project folder, then run these lines one at a time:

```
cd backend  
python -m venv venv  
venv\Scripts\activate  
pip install -r requirements.txt  
python scripts\seed_database.py  
uvicorn backend.main:app --reload --port 8000
```

When you see the line *Application startup complete*, the engine is ready. Keep this window open - closing it stops the engine. You can check it any time by opening <http://localhost:8000/health> in the browser: it should say *healthy*.

Step 3 - Start the website (dashboard)

Open a **second** Command Prompt window and run:

```
cd frontend  
npm install  
npm run dev
```

Then open <http://localhost:5173> in your browser. The dashboard loads the live road-safety map and scores.

Troubleshooting

Tool / Language	What It Is and Why We Use It
Python	The main programming language of the engine. Easy to read and widely used for data work.
FastAPI	A free Python tool that lets the website ask the engine questions and get answers, like a helpful receptionist.
React (with JavaScript)	The toolkit that builds the website screens - the map, lists, panels and buttons - and makes them respond instantly to clicks.
scikit-learn	A machine-learning library: the part that studied past accident records and road features to learn how to score each road piece.

Tool / Language	What It Is and Why We Use It
Leaflet	The free map-drawing library that shows the satellite view of Bengaluru with the coloured road lines.
GeoPandas and OpenStreetMap data	Free world-map data that provides the road layout, junctions, crossings and bus stops used in the safety calculations.
SQLite	A small built-in notebook (database) that stores saved what-if plans and repair projects.
Vite	The tool that starts the website quickly while developers work on it.
HTML and CSS	The standard languages that give the website its structure and its dark control-room styling.

We follow a strict honesty rule: **no accident number is ever invented**. Where information is missing (for example, an unknown speed limit), the system says so instead of guessing, and every road piece carries a confidence label so users know how much to trust each score.

Acknowledgements. We thank the Bengaluru Traffic Police and the OpenCity public data portal for publishing official accident statistics; KGIS and the Karnataka government for police boundary maps; the OpenStreetMap contributors for the free road map; and MoRTH / NCRB for national road-safety research. This project exists to help make Bengaluru's nights safer for everyone.

